

User Experience (UX) Research and Testing Plan for the DashAR System

1. Introduction

User Experience (UX) research and testing are critical in the development of HCI-based systems, ensuring usability and satisfaction in meeting user needs. This UX research plan outlines the methodology for gathering insights, conducting usability testing, and analyzing user interactions with the DashAR system.

Due to time constraints, this plan provides an approach for future researchers to evaluate the UX aspects of this research.

2. Research Objectives

The primary objectives of this UX research are:

- Evaluate the usability of the DashAR system, from system setup to utilization.
- Evaluate how well the HUD interface aligns with user expectations.
- Identify pain points, inefficiencies, and areas of improvement, from system setup to utilization.
- To validate the effectiveness of the HUD interface in real-world scenarios.

3. UX Research Methods

A mixed-method approach combining qualitative and quantitative research is recommended:

3.1. User Interviews and Surveys

- **Objective:** Gather insights into user expectations, needs, and behaviors.
- **Methodology:** Conduct semi-structured interviews with 10–20 representative users and distribute surveys to a broader audience, meeting the following criteria:
 - At least 21 years of age.
 - At least 5 years of experience driving.
 - Possess a valid Driver's License.
- **Tools:**
 - Google Forms (<https://forms.google.com>)
 - Qualtrics (<https://www.qualtrics.com>)
 - Other reputable survey platforms.

3.2. Heuristic Evaluation

- **Objective:** Identify usability issues using Jakob Nielsen's 10 Usability Heuristics.
- **Methodology:** A panel of 3–5 UX experts will review the system and provide structured feedback.
- **Reference:** Nielsen, J. (1994). "10 Usability Heuristics for User Interface Design." NN Group. Available at: <https://www.nngroup.com/articles/ten-usability-heuristics/>

3.3. Usability Testing

- **Objective:** Assess system efficiency, effectiveness, and satisfaction.
- **Methodology:** Conduct usability tests with at least 20 participants using a think-aloud protocol.
- **Metrics Collected:**
 - Task success rate
 - Time on task
 - Error rate
 - System Usability Scale (SUS) score
- **Tools:**
 - Lookback.io (<https://lookback.io>) – for remote usability testing.
 - Morae (<https://www.morae.com/> and https://assets.techsmith.com/Docs/pdf-morae/morae3_datasheet.pdf) – for in-depth usability session recording.

3.4. A/B Testing

- **Objective:** Compare different design variations to determine which performs better.
- **Methodology:** Deploy two versions (A and B) and analyze user engagement metrics.
- **Tools:**
 - VWO (<https://vwo.com>)

3.5. Eye-Tracking and Heatmaps

- **Objective:** Identify areas of user attention and interaction.
- **Methodology:** Utilize eye-tracking software or heatmap analysis.
- **Tools:**
 - Tobii Eye Tracker (<https://www.tobiiipro.com>)
 - Hotjar (<https://www.hotjar.com>)

4. Data Collection and Analysis

- **Qualitative Data:** Thematic analysis of interview transcripts and observational notes.
- **Quantitative Data:** Statistical analysis using Python (Pandas, SciPy) or SPSS.

5. Ethical Considerations

- **Informed Consent:** Participants will provide explicit consent before testing.
- **Data Privacy:** User data will be anonymized and stored securely.
- **Legality:** The use of AR technology while operating an automobile may be of concern. It is recommended to verify the legality of the system's utilization while driving, and potentially consider using a closed loop track for testing.

6. Expected Outcomes

- Insights into system usability and design effectiveness.
- Identification of usability issues and areas for improvement.
- Recommendations for future iterations of the system.

References

- Nielsen, J. (1994). "10 Usability Heuristics for User Interface Design." NN Group. <https://www.nngroup.com/articles/ten-usability-heuristics/>
- Brooke, J. (1996). "SUS: A quick and dirty usability scale." <https://www.usability.gov/how-to-and-tools/methods/system-usability-scale.html>

This UX research and testing plan ensures that the thesis aligns with industry best practices while being practical for graduate research.